Program Summary - A2_Kevin_Lu.sas

Execution Environment

Author:	u64322043
File:	/home/u64322043/BAN130/A2_Kevin_Lu.sas
SAS Platform:	Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host:	ODAWS01-USW2-2.ODA.SAS.COM
SAS Version:	9.04.01M8P02222023
SAS Locale:	en_US
Submission Time:	9/24/2025, 7:11:37 PM
Browser Host:	99.209.173.182
User Agent:	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36
Application Server:	ODAMID00-USW2-2.ODA.SAS.COM

Code: A2_Kevin_Lu.sas

```
* Kevin Lu
* 117926253
* BAN 130
* Assignment Two
* Septmember 26, 2025
*-----------------------------------------------------------------------------------------
* Chatper four Problem 1;
libname learn '/home/u64322043/BAN130';

data learn.Perm;
/* Inputing the variables with formatting */
input ID: $3. Gender: $1. DOB: mmddyy10.
    Height Weight;

    /* Label the input */
    label DOB = 'Date of Birth'
    Height = 'Height in inches'
    Weight = 'Weight in pounds'
    /* Formatting the DOB (Date of Birth) */
    format DOB date9.;

    /* Heres the datasets */
    datalines;
    001 M 10/21/1946 68 150
    002 F 5/26/1950 63 122
    003 M 5/11/1981 72 175
    004 M 7/4/1983 70 128
    005 F 12/25/2005 30 40
    ;

run;
/* Running PROC Contents on this data */
title "Listing ALL the Perm SAS data in a Library";
```

```
proc contents data=learn.Perm;
run;

/* Chapter four Problem 2 */
/* Running Proc Print on this dataset that created in Problem 1 */
title "Listing of this Dataset of Perm";
proc print data=learn.perm;
run;
/* The difference between proc print and viewtable where proc print display
variable name as column headers. For viewtable windows just only showing
labels as colunms headers and making them more descriptive.
/* Chapter five Problem 1 */
/* Creating a dataset as provided */
data voter;
input Age Party : $1. (Ques1-Ques4)($1. + 1);
datalines;
23 D 1 1 2 2
45 R 5 5 4 1
67 D 2 4 3 3
39 R 4 4 4 4
19 D 2 1 2 1
75 D 3 3 2 3
57 R 4 3 4 4
;
run;

/* Create a Formatting using formatting procedure */
proc format;
    value Age 0-30 = '(0-30)'
              31-50 = '(31-50)'
              51-70 = '(51-70)'
              71-high = '(71-Higher)';

    value $Party 'D' = 'Democrat'
                 'R' = 'Republican';

    value Ques 1 = 'Strongly Disagree'
               2 = 'Disagree'
               3 = 'No Opinion'
               4 = 'Agree'
               5 = 'Strongly Agree';
run;
/* Using the Proc Print to list the observation in this data set
Using with label option to show the label */
proc print data = voter label;
title 'Tables for Question 5 Problem 1';
format Age age.
       Party  $Party.
       Ques1-Ques4 $likert.;

label Ques1 = 'The president is doing a good job'
      Ques2 = 'Congress is doing good job'
      Ques3 = 'Taxes are too high'
      Ques4 = 'Government should cut spending';
```

```
run;
/* Using Proc Freq to list frequencies for the four questions */
proc freq data = voter;
title 'Question Frequencies';
    tables Ques1-Ques4 / nocum;
run;

/* Chapter 5 Problem 2
Wanted to use only three categories like:
Generally Disagree (Strongly Disagree + Disagree)
No Opinion
Generally Agree (Agree + Strongly Agree)*/

proc format;
    value $three '1', '2' = 'Generally Disagree'
                 '3' = 'No Opinion'
                 '4', '5' = 'Generally Agree';
run;

/* Applying the nre format to several variables with Proc Freq */
proc freq data = voter;
    title 'Question Frequencies Using Three Categories';
    tables Ques1-Ques4;
```

Log: A2_Kevin_Lu.sas

Notes (24)

```
1          OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69         * Kevin Lu
70         * 117926253
71         * BAN 130
72         * Assignment Two
73         * Septmember 26, 2025
74         *-------------------------------------------------------------------------------------
75         * Chatper four Problem 1;
76         libname learn '/home/u64322043/BAN130';
NOTE: Libref LEARN was successfully assigned as follows:
      Engine:        V9
      Physical Name: /home/u64322043/BAN130
77
78         data learn.Perm;
79         /* Inputing the variables with formatting */
80         input ID: $3. Gender: $1. DOB: mmddyy10.
81         Height Weight;
82
83         /* Label the input */
84         label DOB = 'Date of Birth'
85         Height = 'Height in inches'
86         Weight = 'Weight in pounds'
87         /* Formatting the DOB (Date of Birth) */
88         format DOB date9.;
89
90         /* Heres the datasets */
91         datalines;
```

```
NOTE: The data set LEARN.PERM has 5 observations and 5 variables.
NOTE: DATA statement used (Total process time):
      real time           0.16 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              698.75k
      OS Memory           21668.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        259  Switch Count  1
      Page Faults                       0
      Page Reclaims                     131
      Page Swaps                        0
      Voluntary Context Switches        44
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           272


97         ;
98
99         run;
100        /* Running PROC Contents on this data */
101        title "Listing ALL the Perm SAS data in a Library";
102        proc contents data=learn.Perm;
103        run;

NOTE: PROCEDURE CONTENTS used (Total process time):
      real time           0.17 seconds
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
      memory              1923.53k
      OS Memory           22440.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        260  Switch Count  0
      Page Faults                       0
      Page Reclaims                     211
      Page Swaps                        0
      Voluntary Context Switches        13
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           16


104
105        /* Chapter four Problem 2 */
106        /* Running Proc Print on this dataset that created in Problem 1 */
107        title "Listing of this Dataset of Perm";
108        proc print data=learn.perm;
109        run;

NOTE: There were 5 observations read from the data set LEARN.PERM.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.11 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              704.59k
      OS Memory           22180.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        261  Switch Count  0
      Page Faults                       0
```

```
      Page Reclaims                     75
      Page Swaps                        0
      Voluntary Context Switches        13
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           0


      /* The difference between proc print and viewtable where proc print display
      variable name as column headers. For viewtable windows just only showing
      labels as colunms headers and making them more descriptive.
      /* Chapter five Problem 1 */
      /* Creating a dataset as provided */
      data voter;
      input Age Party : $1. (Ques1-Ques4)($1. + 1);
      datalines;

NOTE: The data set WORK.VOTER has 7 observations and 6 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              670.15k
      OS Memory           22436.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        262  Switch Count  2
      Page Faults                       0
      Page Reclaims                     89
      Page Swaps                        0
      Voluntary Context Switches        17
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           264


      ;
      run;

      /* Create a Formatting using formatting procedure */
      proc format;

    !  value Age 0-30 = '(0-30)'
        31-50 = '(31-50)'
        51-70 = '(51-70)'
        71-high = '(71-Higher)';
NOTE: Format AGE is already on the library WORK.FORMATS.
NOTE: Format AGE has been output.


    !  value $Party 'D' = 'Democrat'
      'R' = 'Republican';
NOTE: Format $PARTY is already on the library WORK.FORMATS.
NOTE: Format $PARTY has been output.


    !  value Ques 1 = 'Strongly Disagree'
         2 = 'Disagree'
         3 = 'No Opinion'
         4 = 'Agree'
         5 = 'Strongly Agree';
NOTE: Format QUES is already on the library WORK.FORMATS.
```

```
NOTE: Format QUES has been output.
143        run;

NOTE: PROCEDURE FORMAT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              297.65k
      OS Memory           22176.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        263  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           32


144        /* Using the Proc Print to list the observation in this data set
145        Using with label option to show the label */
146        proc print data = voter label;
147        title 'Tables for Question 5 Problem 1';
148        format Age age.
149           Party  $Party.
150           Ques1-Ques4 $likert.;
151
152        label Ques1 = 'The president is doing a good job'
153          Ques2 = 'Congress is doing good job'
154          Ques3 = 'Taxes are too high'
155          Ques4 = 'Government should cut spending';
156        run;

NOTE: There were 7 observations read from the data set WORK.VOTER.
NOTE: PROCEDURE PRINT used (Total process time):
      real time           0.01 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              729.09k
      OS Memory           22436.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        264  Switch Count  1
      Page Faults                       0
      Page Reclaims                     67
      Page Swaps                        0
      Voluntary Context Switches        11
      Involuntary Context Switches      0
      Block Input Operations            0
      Block Output Operations           16


157        /* Using Proc Freq to list frequencies for the four questions */
158        proc freq data = voter;
159        title 'Question Frequencies';
160        tables Ques1-Ques4 / nocum;
161        run;

NOTE: There were 7 observations read from the data set WORK.VOTER.
NOTE: PROCEDURE FREQ used (Total process time):
      real time           0.10 seconds
```

```
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
      memory              911.53k
      OS Memory           22696.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        265  Switch Count  3
      Page Faults                       0
      Page Reclaims                     126
      Page Swaps                        0
      Voluntary Context Switches        34
      Involuntary Context Switches      2
      Block Input Operations            0
      Block Output Operations           264


162
163        /* Chapter 5 Problem 2
164        Wanted to use only three categories like:
165        Generally Disagree (Strongly Disagree + Disagree)
166        No Opinion
167        Generally Agree (Agree + Strongly Agree)*/
168
169        proc format;
170
170      !  value $three '1', '2' = 'Generally Disagree'
171         '3' = 'No Opinion'
172         '4', '5' = 'Generally Agree';
NOTE: Format $THREE is already on the library WORK.FORMATS.
NOTE: Format $THREE has been output.
173        run;

NOTE: PROCEDURE FORMAT used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              240.56k
      OS Memory           22176.00k
      Timestamp           09/25/2025 02:11:37 AM
      Step Count                        266  Switch Count  0
      Page Faults                       0
      Page Reclaims                     14
      Page Swaps                        0
      Voluntary Context Switches        0
      Involuntary Context Switches      1
      Block Input Operations            0
      Block Output Operations           0


174
175        /* Applying the nre format to several variables with Proc Freq */
176        proc freq data = voter;
177        title 'Question Frequencies Using Three Categories';
178        tables Ques1-Ques4;
179        format Ques1-Ques4 $three.;
180        run;

NOTE: There were 7 observations read from the data set WORK.VOTER.
NOTE: PROCEDURE FREQ used (Total process time):
      real time           0.08 seconds
      user cpu time       0.03 seconds
      system cpu time     0.00 seconds
```

```
memory                  972.21k
OS Memory               22696.00k
Timestamp               09/25/2025 02:11:37 AM
Step Count                          267  Switch Count  3
Page Faults                         0
Page Reclaims                       120
Page Swaps                          0
Voluntary Context Switches          25
Involuntary Context Switches        3
Block Input Operations              0
Block Output Operations             280
```

```
181
182
183
184        OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
194
```

Results: A2_Kevin_Lu.sas

Listing ALL the Perm SAS data in a Library

The CONTENTS Procedure

Data Set Name	LEARN.PERM	**Observations**	5
Member Type	DATA	**Variables**	5
Engine	V9	**Indexes**	0
Created	09/24/2025 19:11:37	**Observation Length**	32
Last Modified	09/24/2025 19:11:37	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	4078
Obs in First Data Page	5
Number of Data Set Repairs	0
Filename	/home/u64322043/BAN130/perm.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	12415142115
Access Permission	rw-r--r--
Owner Name	u64322043
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Label
3	DOB	Num	8	Date of Birth
2	Gender	Char	1	
4	Height	Num	8	Height in inches
1	ID	Char	3	

Alphabetic List of Variables and Attributes				
#	Variable	Type	Len	Label
5	Weight	Num	8	Weight in pounds format DOB date9.

Listing of this Dataset of Perm

Obs	ID	Gender	DOB	Height	Weight
1	00	M	-4820	68	150
2	00	F	-3507	63	122
3	00	M	7801	72	175
4	00	M	8585	70	128
5	00	F	16795	30	40

Tables for Question 5 Problem 1

Obs	Age	Party	The president is doing a good job	Congress is doing good job	Taxes are too high	Government should cut spending
1	(0-30)	Democrat	1	1	2	2
2	(31-50)	Republican	5	5	4	1
3	(51-70)	Democrat	2	4	3	3
4	(31-50)	Republican	4	4	4	4
5	(0-30)	Democrat	2	1	2	1
6	(71-Higher)	Democrat	3	3	2	3
7	(51-70)	Republican	4	3	4	4

Question Frequencies

The FREQ Procedure

Ques1	Frequency	Percent
1	1	14.29
2	2	28.57
3	1	14.29
4	2	28.57
5	1	14.29

Ques2	Frequency	Percent
1	2	28.57
3	2	28.57
4	2	28.57
5	1	14.29

Ques3	Frequency	Percent
2	3	42.86
3	1	14.29
4	3	42.86

Ques4	Frequency	Percent
1	2	28.57
2	1	14.29
3	2	28.57

Ques4	Frequency	Percent
4	2	28.57

Question Frequencies Using Three Categories

The FREQ Procedure

Ques1	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Generally Disagree	3	42.86	3	42.86
No Opinion	1	14.29	4	57.14
Generally Agree	3	42.86	7	100.00

Ques2	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Generally Disagree	2	28.57	2	28.57
No Opinion	2	28.57	4	57.14
Generally Agree	3	42.86	7	100.00

Ques3	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Generally Disagree	3	42.86	3	42.86
No Opinion	1	14.29	4	57.14
Generally Agree	3	42.86	7	100.00

Ques4	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Generally Disagree	3	42.86	3	42.86
No Opinion	2	28.57	5	71.43
Generally Agree	2	28.57	7	100.00